

Age, Sex, Consumption and Cancer: A Support Vector Machine Approach

DATA 5322

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INTRODUCTION

This study, based on data from the 2022 National Health Interview Survey, uses Support Vector Machines (SVMs) to predict cancer presence based on demographic (age, sex) and consumption factors (sweetened coffee, juice, sport drinks, alcohol).

THEORETICAL BACKGROUND

SVMs classify data by finding an optimal decision boundary (hyperplane) that maximizes separation between classes. Performance evaluated using *accuracy, precision, recall, F1-score and ROC-AUC*. The soft margin optimization formula:

$$\min_{\mathbf{w}, \mathbf{b}, \xi} \left(\frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n \xi_i \right)$$

where $\mathbf{w}$ is the orientation of the hyperplane, $\mathbf{b}$ shifts the hyperplane, $\mathbf{C}$ is the tradeoff between and errors.

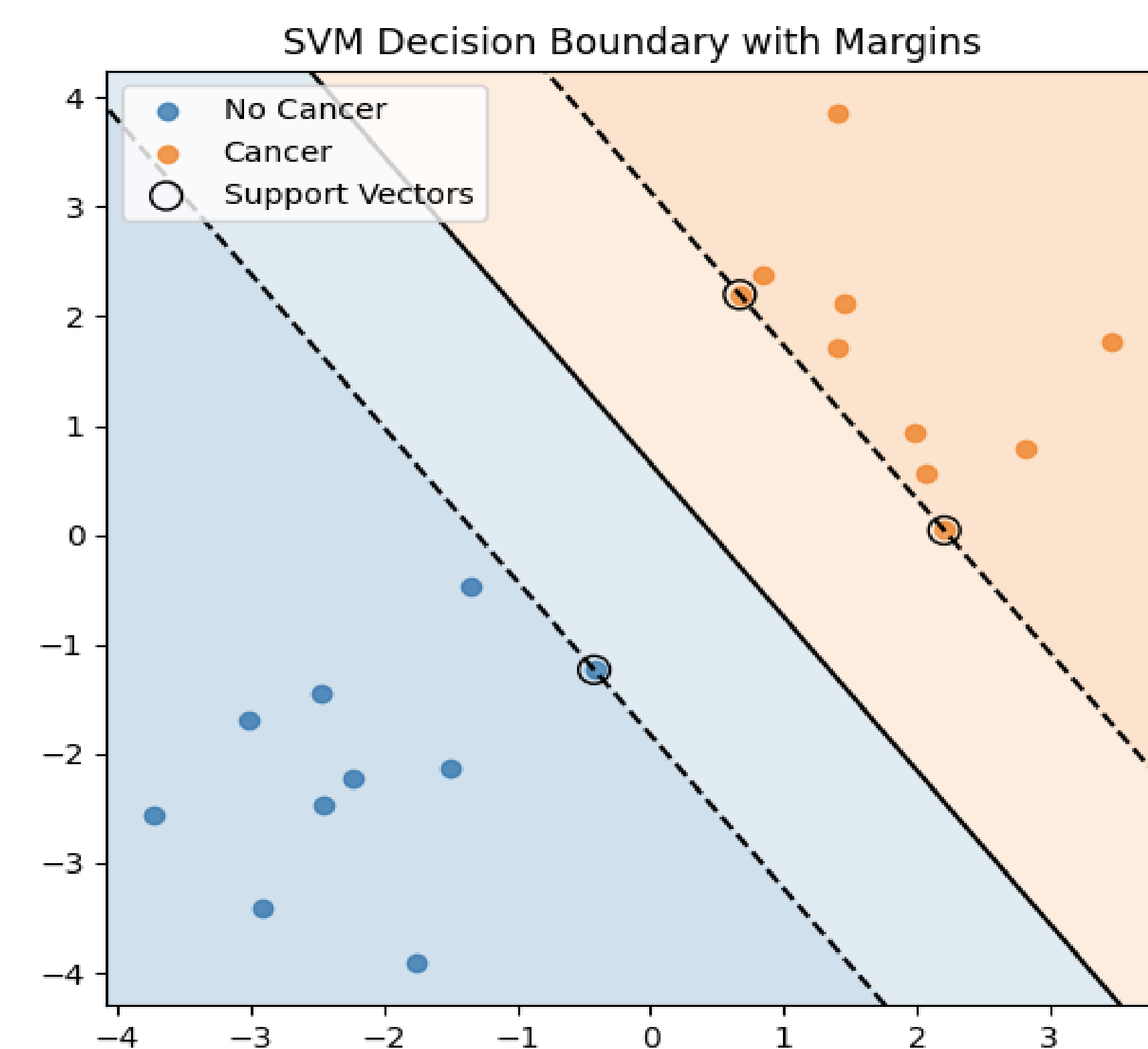


Figure 1. Diagram of sample decision boundary plot

METHODOLOGY

Preprocessing: median imputation for missing predictors and feature standardization.

Modeling: Train/test split (70/30, stratified) with linear, polynomial, and RBF kernels

Hyperparameter Tuning:

- GridSearchCV: cross-validation of 5 folds.
- Class imbalance handled using class weights

Parameters tested:

- C: [0.1, 1, 10]
- gamma: ['scale']
- degree: [2, 3]

RESULTS

	Linear	Poly	Radial
Train accuracy	0.665	0.679	0.686
Test accuracy	0.671	0.679	0.660
Precision (Cancer)	0.25	0.25	0.24
Recall (Cancer)	0.81	0.79	0.81
F1-score (Cancer)	0.38	0.38	0.37
ROC-AUC	0.787	0.781	0.776

Figure 2. Performance result table of cancer cases for linear, polynomial and radial kernel

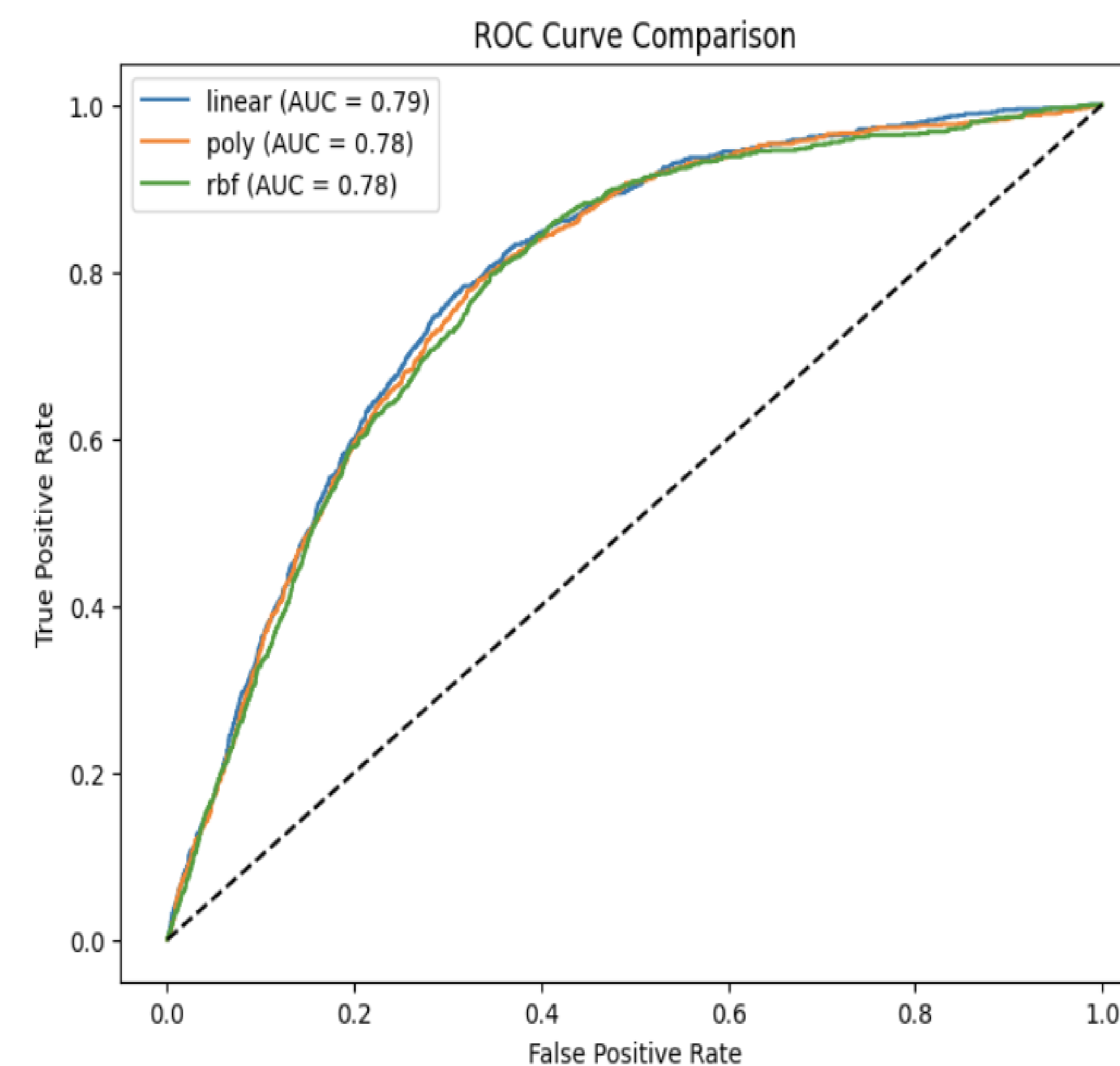


Figure 3. ROC curve comparison for linear, polynomial and radial kernel

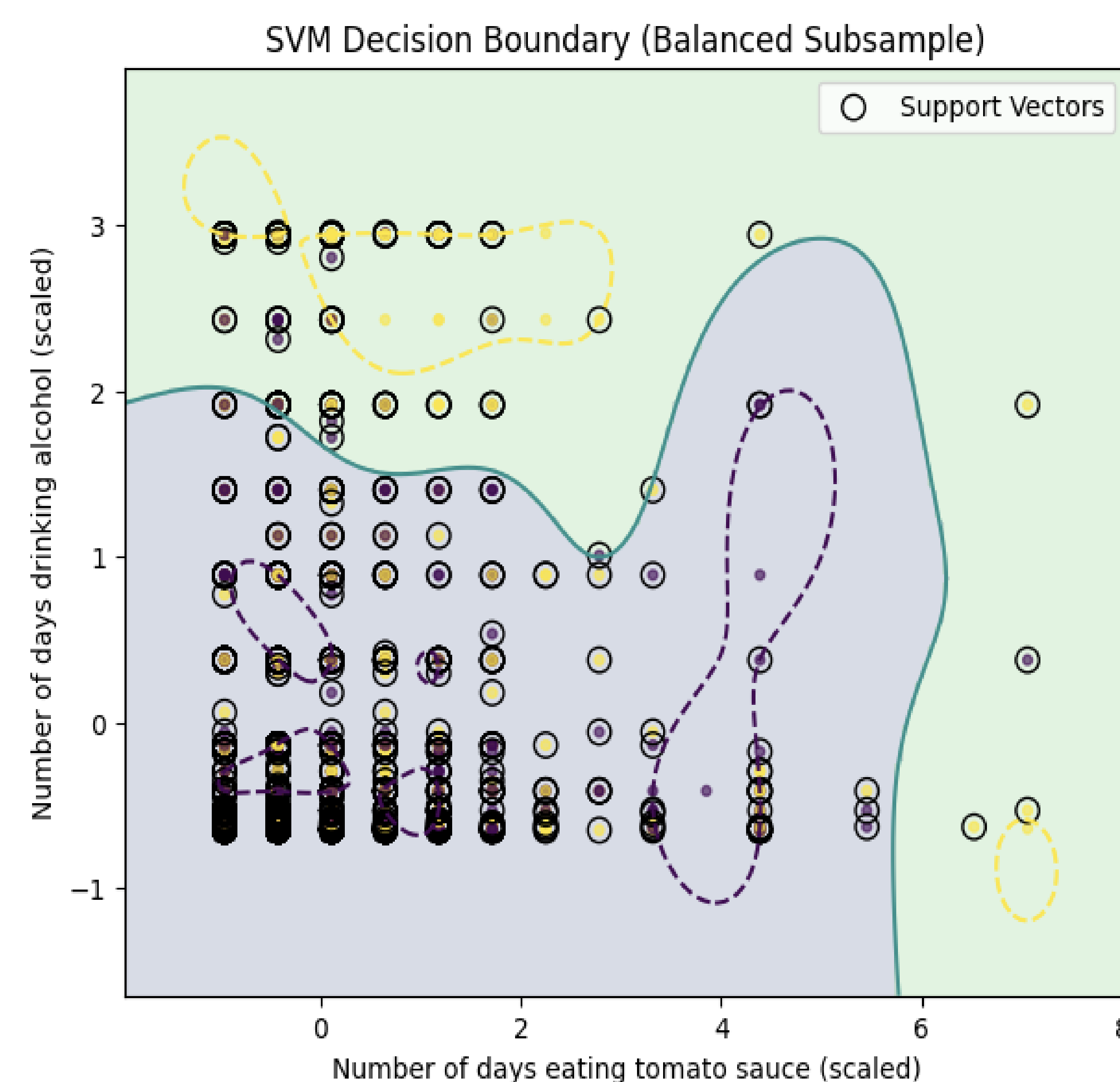


Figure 4. Decision boundary plot with balanced subsample

KEY FINDINGS

- All models performed similarly in accuracy and ROC-AUC
- Polynomial kernel achieved highest accuracy (0.679)
- Linear kernel achieved highest ROC-AUC (0.787)
- High recall for cancer cases (0.80)
- Low precision (0.25) → many false positives

Feature importance:

- Age dominates prediction (0.85)
- Minor contributions from: tomato sauce (0.10), sex (0.05), alcohol & sports drinks (~0.02)

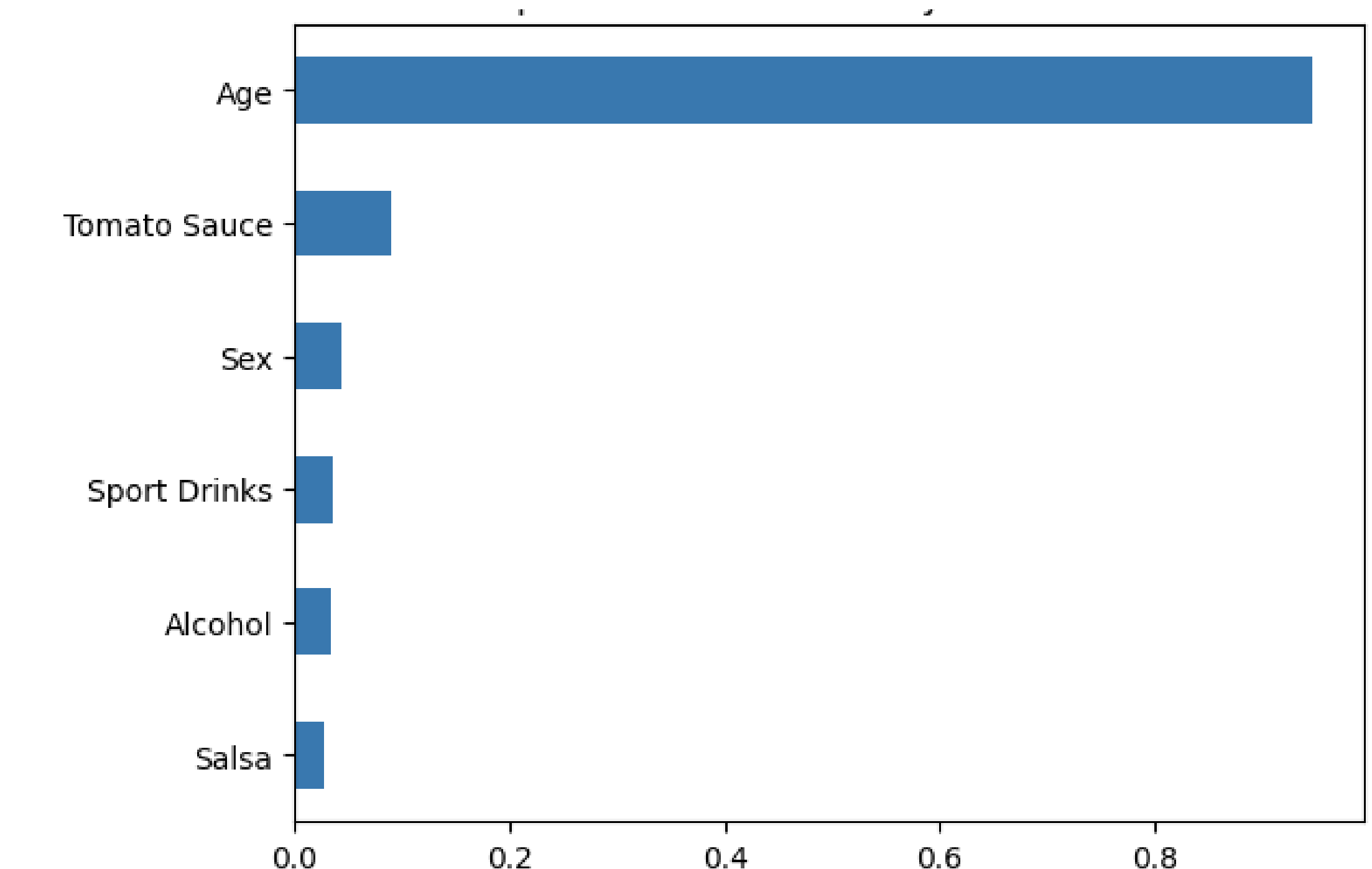


Figure 5. Feature Importance Plot

DISCUSSION

Model performance was consistent across kernels, suggesting limited nonlinear structure in the data. High recall but low precision indicates the model identifies most cancer cases but over-predicts risk. The model results suggest that age is the dominant predictor of cancer risk, with diet factors contributing only marginally. This reinforces the importance of age-targeted screening and early detection strategies.

CONCLUSION

SVMs show moderate ability to predict cancer using survey data. Results highlight the dominant role of age in disease prediction. Behavioral variables alone provide limited predictive power.

Future work:

- Improve feature set
- Address class imbalance more aggressively
- Explore alternative models (e.g., ensemble methods)

REFERENCES

Lynn A. Blewett, Julia A. Rivera Drew, Miriam L. King, Kari C.W. Williams, Daniel Backman, Annie Chen, and Stephanie Richards. IPUMS Health Surveys: National Health Interview Survey, Version 7.4 [dataset]. Minneapolis, MN: IPUMS, 2024. <https://doi.org/10.18128/D070.V7.4>

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